American College of Technology (ACT)

Department of Computer Science

RESEARCH SYNOPSIS
Adversarial Attacks and Defenses in Machine Learning-Based Financial Fraud Detection

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Background and Rationale

Financial fraud is a persistent cybersecurity and financial-sector problem because fraudulent transactions can cause direct financial loss, undermine customer trust, and increase operational risk. Machine learning has become an important approach for identifying suspicious transactions because models can learn patterns from large transaction datasets and support automated fraud screening. However, the security of the fraud-detection model itself is an important concern. An attacker may deliberately manipulate selected transaction attributes so that a fraudulent transaction appears legitimate to the model. Such deliberately crafted inputs are commonly referred to as adversarial examples.

This research focuses on the security and robustness of machine learning-based financial fraud detection. Rather than considering fraud detection only as a prediction problem, the study treats the model as a cybersecurity asset that may be targeted by an adversary. The research will examine how adversarial manipulation can reduce the reliability of fraud detection and investigate practical defensive techniques that can make the model more resistant to such manipulation.

Problem Statement

Conventional machine learning fraud detectors are generally evaluated under normal data conditions, where the input is assumed to be genuine and correctly represented. In a real cybersecurity environment, however, an attacker may intentionally alter input characteristics to evade detection. A model that performs well on ordinary test data can therefore become less reliable when exposed to carefully crafted adversarial transactions. This creates a security gap between predictive performance and operational robustness.

For a cybersecurity student, the problem is particularly relevant because it connects machine learning with adversarial security, threat modeling, and defensive mechanisms. There is a need for a practical experimental study that demonstrates the vulnerability of financial fraud classifiers to adversarial manipulation and evaluates whether suitable defenses can improve their resilience without making the research dependent on expensive computing resources.

Research Gap and Proposed Focus

Many fraud-detection studies concentrate primarily on improving classification performance under normal conditions. Comparatively less attention is given to the security of the trained model when an attacker intentionally manipulates transaction features. This study addresses that gap by combining financial fraud detection with adversarial machine learning. The central focus is to establish a baseline fraud-detection model, expose it to realistic adversarial manipulation for tabular transaction data, and then investigate defensive strategies designed to reduce the resulting vulnerability.

The proposed experimental approach is designed to be reproducible and suitable for an academic research environment.

Research Concept

The research follows a security-oriented experimental framework. First, a conventional machine learning fraud detector will be developed using financial transaction data. The baseline establishes how the model behaves under normal conditions. Second, adversarial inputs will be generated by making controlled changes to appropriate transaction features while maintaining the practical constraints of tabular financial data. The purpose is to determine whether fraudulent transactions can be modified in ways that cause the detector to classify them incorrectly.

After the vulnerability has been demonstrated, defensive approaches will be applied to improve model robustness. The study will investigate adversarially informed training and related model-hardening techniques, then compare the behavior of the original and defended systems under both ordinary and adversarial conditions. The comparison will focus on whether the defense reduces successful evasion while preserving useful fraud-detection capability.

Significance of the Study

The study is significant from a cybersecurity perspective because financial fraud models can themselves become targets of attack. Understanding this threat can help researchers and practitioners think beyond the traditional question of whether a model is accurate and consider whether it remains dependable when deliberately targeted. The work can also provide a practical demonstration of adversarial machine learning using commonly available computing resources.

Academically, the research contributes a focused connection between cybersecurity, machine learning, and financial fraud detection. Practically, it can provide evidence about common weaknesses of fraud classifiers and the value of defensive techniques. The resulting methodology can also serve as a foundation for future research on robust machine learning in other security-sensitive domains.

Expected Outcome

The expected outcome is an experimentally supported understanding of how adversarial manipulation affects machine learning-based financial fraud detection and how defensive techniques can improve model resilience. The study is intended to produce a reproducible experimental workflow covering data preparation, baseline model development, adversarial testing, defense implementation, and comparative analysis. The use of CPU-feasible machine learning methods makes the study appropriate for a postgraduate cybersecurity research project with limited computing resources.

Scope and Limitation

The study will focus on machine learning-based financial transaction fraud detection and adversarial manipulation of tabular input data. It will use appropriately prepared financial transaction data rather than confidential institutional banking records. The research will evaluate model behavior in a controlled experimental environment and will not attempt to reproduce every operational control used by financial institutions. The work will therefore provide an academic assessment of model robustness rather than a complete production banking security system.

Keywords

Financial Fraud Detection; Machine Learning; Adversarial Attacks; Adversarial Defense; Cybersecurity; Model Robustness